

Comments on “Rational Priors for Imprecise Credence”

The question: in an *imprecise-credences* framework, in which epistemic states are represented as sets of credence functions, what are the rational epistemic states for an agent with no evidence?

The results: Suppose we adopt an inaccuracy measure for imprecise credences that includes an “informativity” parameter α that gives relative weighting to the most- and least-accurate credence functions in a set.

1. Wang characterizes the class of credal sets that minimize maximal expected inaccuracy for any given value of α , exemplified by the **natural sharpenings**. These are the rational credal sets according to a **maximin** rule.
2. Wang shows that the fully-imprecise credal set and the uniform distribution are both accuracy-dominated.

Different dimensions of epistemic risk-sensitivity

Wang’s imprecise inaccuracy measure involves an “informativity” parameter:

Definition (Definition 2: Imprecise Inaccuracy Measure). For any credal set C and any world $w \in W$, the inaccuracy of C at w , given informativity parameter $\alpha \in [0, 1]$, is given by:

$$I^\alpha(C, w) = \alpha \min_{c \in C} I(c, w) + (1 - \alpha) \max_{c \in C} I(c, w).$$

Wang writes that, intuitively, the informativity parameter α can represent an agent’s attitude toward epistemic risk.¹

Another risk-sensitive apparatus occurs in the Maximin rationality constraint:

Constraint (Maximin). A credal set X is irrational (relative to informativity parameter α) if there exists a credal set C such that both of the following hold:

- (i) $\max_{w \in W} I^\alpha(C, w) < \max_{w \in W} I^\alpha(X, w)$
- (ii) There exists no credal set $\mathcal{Y}$ such that $\max_{w \in W} I^\alpha(\mathcal{Y}, w) < \max_{w \in W} I^\alpha(C, w)$

Maximin is risk averse in that it evaluates the rationality of credal states based on the maximum inaccuracy they attain over the possible worlds.²

Wang writes that these are two different types of risk aversion, and that they “are orthogonal, even in a literal geometric sense.”

Question 1: How do the two formal implementations of risk parameters connect to intuitive considerations of risk-sensitivity in epistemic decision theory? If so, how do the two notions of risk intuitively differ?

Two possible answers: (1) there is a single notion of risk, implemented both with α and Maximin, and a more risk-averse agent should implement the risk-aversion in both places. (2) There are two notions of risk, which are respectively implemented with the informativity parameter and Maximin. One can be risk-seeking relative to one notion and risk-averse relative to the other.

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I repeat Wang’s definition of natural sharpening and her main results (Theorems 1 and 2) on the last page of this handout. I don’t include discussion of the dominance result, for space reasons.

We superscript the imprecise inaccuracy measures with the informativity parameter. This is to help us keep clear that

1. $I : \mathcal{P} \times W \rightarrow [0, \infty]$ is a precise inaccuracy measure and
2. Each $I^\alpha : 2^{\mathcal{P}} \times W \rightarrow [0, \infty]$ is an imprecise inaccuracy measure with informativity parameter α , defined in turns of I .

¹ **Wang:** “one might hold that a more spread-out credal set—one that allows widely dispersed probability functions—is more risk-averse than a more tightly clustered credal set. This intuition concerns the degree of epistemic commitment or informativeness of the credal state itself. ... this intuition is captured by the informativity parameter α : larger values of α favor less informative, more internally diverse credal sets, whereas smaller values favor more informative, narrowly concentrated credal sets.”

² **Wang:** “one might hold that a credal set whose inaccuracy is more symmetrically distributed across possible worlds is more risk-averse than one whose inaccuracy is concentrated on a smaller range of worlds. This intuition is naturally captured by the Maximin decision rule. Maximin recommends credal sets that minimize their maximum inaccuracy across worlds. As a result, if the inaccuracy profile of one credal set is more evenly spread across possible worlds than that of another, the former will be preferred by Maximin, even if both perform similarly on average. In this sense, Maximin favors credal sets that avoid large epistemic losses at any particular world.”

Characterizing non-extreme attitudes to risk

Suppose there are two distinct senses of risk: **a-risk**, encoded into the informativity parameter α , and **b-risk**, implemented in the rationality constraint. Then Wang's epistemic agents have variable aversion to a-risk, but are always maximally averse to b-risk!

The Hurwicz criterion, as implemented in Konek (forthcoming) allows us to vary aversion to b-risk as well.

Definition. The Hurwicz-inaccuracy of a credal set C relative to informativity parameter α and b-risk parameter β is given by

$$\mathcal{H}^\beta I^\alpha(C) = \beta \max_{w \in W} I^\alpha(C, w) + (1 - \beta) \min_{w \in W} I^\alpha(C, w)$$

We can use Hurwicz-inaccuracy to generalize Maximin:

Constraint (Hurwicz). A credal set X is irrational (relative to informativity parameter α and risk parameter β) if there exists a credal set C such that both of the following hold:

- (i) $\mathcal{H}^\beta I^\alpha(C) < \mathcal{H}^\beta I^\alpha(X)$
- (ii) There exists no credal set $\mathcal{Y}$ such that $\mathcal{H}^\beta I^\alpha(\mathcal{Y}) < \mathcal{H}^\beta I^\alpha(C)$

When $\beta = 1$, this is Maximin. As $\beta \rightarrow 0$, more weight is given to the best case world. At $\beta = 0$, we get

Constraint (Minimin). A credal set X is irrational (relative to informativity parameter α) if there exists a credal set C such that both of the following hold:

- (i) $\min_{w \in W} I^\alpha(C, w) < \min_{w \in W} I^\alpha(X, w)$
- (ii) There exists no credal set $\mathcal{Y}$ such that $\min_{w \in W} I^\alpha(\mathcal{Y}, w) < \min_{w \in W} I^\alpha(C, w)$

With Hurwicz inaccuracy in mind, I have a second question.

Question 2: Is there interest in characterizing the rationally permissible sets where the a-risk/informativity parameter is non-extreme, but the b-risk/Hurwicz parameter is extreme? What justifies keeping the b-risk parameter extreme while varying the a-risk parameter?

Hurwicz behavior

I'll end with some discussion about Question 2, by way of some generalization of the Maximin results to Hurwicz. Konek (fc) gives a limited characterization of Hurwicz behavior to show that conservatism in both parameters speaks against the Principle of Indifference:

Theorem (Konek (fc)). If our epistemic utility measure and our Hurwicz rule are both conservative, in the sense that $\alpha > 1/2$ and $\beta > 1/2$ — i.e., they place more of an emphasis on avoiding error/worst-case catastrophe than on positively pinning down the truth/maximising best-case spoils — then there is some imprecise credal state C that our Hurwicz rule recommends over all precise credal states.³

Take the maximum inaccuracy over worlds and weight it by β , then take the minimum inaccuracy over worlds and weight it by $1 - \beta$.

For both α and β , a higher value indicates more risk aversion.

³ In fact, the credal state C that Konek uses is the natural sharpening C^α .)

As a start to considering the general behavior, here are two propositions. First, suppose we set $\beta = 0$, so that the rule is Minimin. Then any maximally-opinionated credal set is permissible:

Proposition 1. *Among possible credal sets, the quantity $\min_{w \in W} I^\alpha(C, w)$ always attains a minimum at $C = \{\mathbb{1}_{w_0}\}$, where $w_0 \in W$ is an arbitrary world. That is, for any $w_0 \in W$,*

$$\{\mathbb{1}_{w_0}\} \in \arg \min_{C \in 2^{\mathcal{P}}} [\min_{w \in W} I^\alpha(C, w)].$$

Minimin puts all the weight on the best-case world, where inaccuracy is minimized by the maximally-committal and maximally-correct credal state, which is the singleton set of the truth-function of the actual world.⁴

A second proposition suggests that for intermediate values of β , the natural sharpenings might not always be rationally permissible, because there are other credal states with lower Hurwicz scores.

Proposition 2. *Let X be the convex hull of $C^\alpha \cup \{c_i^\gamma\}$, where $\gamma \in (\alpha, 1]$. Then*

$$\mathcal{H}^\beta I^\alpha(X) < \mathcal{H}^\beta I^\alpha(C^\alpha)$$

if and only if

$$\frac{I(c_i^\gamma, w_j) - I(c_i^\alpha, w_j)}{I(c_i^\alpha, w_i) - I(c_i^\gamma, w_i)} \cdot \frac{1 - \alpha}{\alpha} < \frac{1 - \beta}{\beta}.$$

Here $I(c_i^\gamma, w_j) - I(c_i^\alpha, w_j)$ is the worst-case difference in accuracy between c_i^γ and c_i^α , and $I(c_i^\alpha, w_i) - I(c_i^\gamma, w_i)$ is the best-case difference in accuracy⁵.

$$\frac{\text{worst-case accuracy difference}}{\text{best-case accuracy difference}} \cdot \text{a-risk seekingness} < \text{b-risk seekingness}$$

For a fixed α , as β increases, the inaccuracy difference ratio must decrease, so γ must decrease and approach α , and if $\beta = 1$ the inequality cannot be satisfied. For a fixed β , as α increases, the inequality can be satisfied with larger γ . So the Hurwicz criterion favors credal sets that involve some gambles, represented by opinionated credal sets $\{c_i^\gamma\}$.⁶

Also, $\mathcal{H}^\beta I^\alpha(C^\gamma) > \mathcal{H}^\beta I^\alpha(C^\alpha)$, so smoothing out X by just expanding to a wider natural sharpening won't do.⁷ So more risk-seeking values of β seem to favor spiky, uneven credal sets, while more risk-seeking values of α do not. This may be one reason to think that variable values of α can be rational, but variable values of β might not be, getting us some response to Question 2.

Proof of Proposition 1

Let X be an arbitrary credal set, $\bar{x}_w = \arg \max_{x \in X} I(x, w)$, $\underline{x}_w = \arg \min_{x \in X} I(x, w)$, and $w_1 \in \arg \min_{w \in W} I^\alpha(X, w)$. Then

$$\begin{aligned} \min_{w \in W} I^\alpha(\{\mathbb{1}_{w_0}\}, w) &= I^\alpha(\{\mathbb{1}_{w_0}\}, w_0) = \alpha \min_{c \in \{\mathbb{1}_{w_0}\}} I(c, w_0) + (1 - \alpha) \max_{c \in \{\mathbb{1}_{w_0}\}} I(c, w_0) \\ &= \alpha I(\mathbb{1}_{w_0}, w_0) + (1 - \alpha) I(\mathbb{1}_{w_0}, w_0) \\ &\quad (\text{extensionality}) = \alpha I(\mathbb{1}_{w_1}, w_1) + (1 - \alpha) I(\mathbb{1}_{w_1}, w_1) \\ &\quad (\text{truth directedness}) \leq \alpha I(\underline{x}_{w_1}, w_1) + (1 - \alpha) I(\bar{x}_{w_1}, w_1) = \min_{w \in W} I^\alpha(X, w) \end{aligned}$$

Here $\{\mathbb{1}_{w_0}\}$ is the credal set with only one member: the credence function assigning 1 to w_0 and 0 to every other world.

⁴ When $\beta = 0$, the value of α does not matter.

⁵ where the more inaccurate one is placed first so both differences are positive.

If $\alpha = \gamma = 0.5$, the condition reduces to worst-case accuracy difference < best-case accuracy difference.

⁶ The specific dependence of γ on α and β will depend on the scoring rule in question. But we know from Proposition 2 that it is not C^α , and I conjecture that it is minimized (perhaps by a set of the form of X in Proposition 2).

⁷ We have (as shown in the appendix) $\mathcal{H}^\beta I^\alpha(C^\gamma) = I^\alpha(C^\gamma, w)$, and by Lemma 1 in Wang's paper, $I^\alpha(C^\gamma, w) = \mathbb{E}_{C_i^\alpha} I(c_i^\gamma)$, so $I^\alpha(C^\gamma, w) = \mathbb{E}_{C_i^\alpha} I(c_i^\gamma) > \mathbb{E}_{C_i^\alpha} I(c_i^\alpha) = I^\alpha(C^\alpha, w)$.

Proof of Proposition 2

Let $\mathcal{X}$ be the convex hull of $C^\alpha \cup \{c_i^\gamma\}$, where $\gamma \in (\alpha, 1]$. We want to know the circumstances in which $\mathcal{H}^\beta I^\alpha(C^\alpha) - \mathcal{H}^\beta I^\alpha(\mathcal{X}) > 0$. First, we know that

$$\begin{aligned}\mathcal{H}^\beta I^\alpha(C^\alpha) &= \beta \max_{w \in W} I^\alpha(C^\alpha, w) + (1 - \beta) \min_{w \in W} I^\alpha(C^\alpha, w) \\ &= \beta I^\alpha(C^\alpha, w_j) + (1 - \beta) I^\alpha(C^\alpha, w_i) \\ &= I^\alpha(C^\alpha, w_j) = \alpha I(c_j^\alpha, w_j) + (1 - \alpha) I(c_i^\alpha, w_j) \\ &= \alpha I(c_j^\alpha, w_j) + (1 - \alpha) I(c_j^\alpha, w_i).\end{aligned}$$

Meanwhile,

$$\begin{aligned}\mathcal{H}^\beta I^\alpha(\mathcal{X}) &= \beta \max_{w \in W} I^\alpha(\mathcal{X}, w) + (1 - \beta) \min_{w \in W} I^\alpha(\mathcal{X}, w) \\ &= \beta \max_{w \in W} [\alpha \min_{c \in \mathcal{X}} I^\alpha(c, w) + (1 - \alpha) \max_{c \in \mathcal{X}} I^\alpha(c, w)] \\ &\quad + (1 - \beta) \min_{w \in W} [\alpha \min_{c \in \mathcal{X}} I^\alpha(c, w) + (1 - \alpha) \max_{c \in \mathcal{X}} I^\alpha(c, w)].\end{aligned}$$

Since the accuracy of the $c \in \mathcal{X}$ depends only on whether $w = w_i$, we know that the quantity in brackets is either minimized at w_i by (i) $\alpha I(c_i^\gamma, w_i) + (1 - \alpha) I(c_j^\alpha, w_i)$ and maximized at w_j by (ii) $\alpha I(c_j^\alpha, w_j) + (1 - \alpha) I(c_i^\gamma, w_j)$, or *vice versa*. And since the difference (i) - (ii) is given by

$$\begin{aligned}&\alpha I(c_i^\gamma, w_i) + (1 - \alpha) I(c_j^\alpha, w_i) - [\alpha I(c_j^\alpha, w_j) + (1 - \alpha) I(c_i^\gamma, w_j)] \\ &= \alpha [I(c_i^\gamma, w_i) - I(c_j^\alpha, w_j)] + (1 - \alpha) [I(c_j^\alpha, w_i) - I(c_i^\gamma, w_j)] < 0,\end{aligned}$$

We know that (ii) > (i). Therefore the quantity in brackets is minimized at w_i by (i) and maximized at w_j by (ii). So we have

$$\mathcal{H}^\beta I^\alpha(\mathcal{X}) = \beta [\alpha I(c_j^\alpha, w_j) + (1 - \alpha) I(c_i^\gamma, w_j)] + (1 - \beta) [\alpha I(c_i^\gamma, w_i) + (1 - \alpha) I(c_j^\alpha, w_i)].$$

Thus

$$\begin{aligned}\mathcal{H}^\beta I^\alpha(C^\alpha) - \mathcal{H}^\beta I^\alpha(\mathcal{X}) &= \alpha I(c_j^\alpha, w_j) + (1 - \alpha) I(c_j^\alpha, w_i) \\ &\quad - \beta [\alpha I(c_j^\alpha, w_j) + (1 - \alpha) I(c_i^\gamma, w_j)] \\ &\quad - (1 - \beta) [\alpha I(c_i^\gamma, w_i) + (1 - \alpha) I(c_j^\alpha, w_i)] \\ &= \alpha (1 - \beta) I(c_j^\alpha, w_j) + (1 - \alpha) \beta I(c_j^\alpha, w_i) \\ &\quad - \alpha (1 - \beta) I(c_i^\gamma, w_i) - (1 - \alpha) \beta I(c_i^\gamma, w_j) \\ &= \alpha (1 - \beta) [I(c_j^\alpha, w_i) - I(c_i^\gamma, w_i)] + (1 - \alpha) \beta [I(c_i^\gamma, w_j) - I(c_j^\alpha, w_j)]\end{aligned}$$

We can rewrite this as the condition

$$\frac{I(c_j^\gamma, w_j) - I(c_i^\alpha, w_j)}{I(c_i^\alpha, w_i) - I(c_j^\gamma, w_i)} \cdot \frac{1 - \alpha}{\alpha} < \frac{1 - \beta}{\beta}.$$

Some Definitions and Results Repeated

The two standard views for rational priors:

The Uniform view. At the beginning of an agent's epistemic life, rationality requires that the credal set consist solely of the uniform distribution.

The Full Imprecision view. At the beginning of epistemic life, rationality requires that the credal set include all probability functions.

The full-imprecision view recommends the most spread-out credal set, while the uniform view recommends a "sharp" credal set with only one credence function, the uniform distribution. We can think of intermediate values between full imprecision and the uniform view. From full imprecision, we can shrink the set by throwing out the most confident credence functions until the most opinionated ones are at most $s \in [0, 1]$ confident in any particular world:

Definition (Natural Sharpening). Let $s \in [\frac{1}{n}, 1]$. For any $i \leq n$, let c_i^s be the probability function defined by $c_i^s(X_i) = s$ and $c_i^s(X_j) = \frac{1-s}{n-1}$ for all $j \neq i$. We denote by C^s the convex hull of $c_1^s, \dots, c_n^s$ and call it the **natural sharpening** with widening parameter s .

Wang proves that the natural sharpenings always minimize expected inaccuracy:

Theorem 1 (Maximin). *Given an imprecise inaccuracy measure with informativity α :*

1. Let $\alpha \geq 1/n$. Then for any credal set $\mathcal{X}$,

$$\max_{w \in W} I^\alpha(C^\alpha) \leq \max_{w \in W} I^\alpha(\mathcal{X})$$

2. Let $\alpha \leq 1/n$. Then for any credal set $\mathcal{X} \neq \{u\}$ (where u is the uniform distribution),

$$\max_{w \in W} I^\alpha(\{u\}) \leq \max_{w \in W} I^\alpha(\mathcal{X})$$

Given the maximin rationality constraint, Theorem 1 entails that the natural sharpenings are rationally permissible for $\alpha \geq 1/n$.

Wang also shows that given the maximin constraint, the rationally permissible credal sets are those whose most accurate member at any w_j is c_i^α and whose least accurate member at any w_i attains inaccuracy $I(c_j^\alpha, w_i)$:

Theorem 2 (Characterization of Rational Sharpening). *Let $\alpha > 1/n$. Suppose C is a credal set, $c_w \in \arg \min_{c \in C} I(c, w)$, and $\bar{c}_w \in \arg \max_{c \in C} I(c, w)$. Then the following are equivalent:*

1. (a) $c_{w_i} = c_i^\alpha$ for all $i \leq n$
(b) $I(\bar{c}_{w_i}, w_i) = I(c_j^\alpha, w_i)$ for all $i \leq n$ and $j \neq i$
2. For any credal set $\mathcal{X}$,

$$\max_{w \in W} I^\alpha(C^\alpha) \leq \max_{w \in W} I^\alpha(\mathcal{X})$$